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Reinforcement Learning with Artificial neural network

Lecture 1

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Outline

- **Machine Learning Overview**
- **Reinforcement learning**
- **Overview about Artificial Neural Networks**

Machine Learning Overview

The possibility of assigning tasks to machines to avoid repetitive actions as a theoretical formulation has been postulated through the ages, and it is recurring in our collective imagination.

Nowadays, artificial intelligence (AI) has found application in many areas including automated customer support and tailoring, advanced driver assistance systems, etc.

ML borrows its foundations from the concepts of early AI research, and its approach to problem-solving is focused on statistical knowledge.

ML systems can be classified in broad categories based on:

- **Whether or not they are trained with human supervision (supervised, unsupervised, semi-supervised, and reinforcement learning);**
- **Whether or not they can learn with new data on the fly (online vs. batch learning);**
- **Whether they only compare new data points to already known data points or if they learn patterns present in the data (instance-based vs. model-based training).**

Supervised or Unsupervised Learning

In supervised learning, the data that are fed to the system include the desired solution, called label.

A classification problem is a typical example of this type of system since the algorithm is given data points that are classified with the correct label and it learns the statistical properties or patterns in those data points to achieve said label.

A regression problem can also be solved using a supervised learning system, since the objective is to predict a numeric value, given a set of features (input variables).

To train these systems, one must feed the algorithm with data points which each contains a set of features and the numeric value it must predict.

The system then learns the relation between the output and the set of features.

Some of the more important supervised learning algorithms are linear regression, logistic regression, polynomial regression, decision trees, SVMs (Fig. 1a), and ANNs.

In unsupervised learning, the data used to train the system are unlabeled. These algorithms form groups of data points based on their features without attributing a class to each group since there are no labels.

This is the main focus of clustering and visualization algorithms. Dimensionality reduction is a related task to these systems, and its objective is to simplify the data while keeping the information by merging features that are strongly correlated.

This method is called feature extraction, and it is used to reduce the input dimensions and make it easier to visualize the grouped data points.

These systems are useful for anomaly detection since it groups the "normal" data points.

If a new data point stays far away from the "normal" group, it might be an anomaly.

Some of the more important unsupervised learning algorithms are clustering algorithms like k-means and visualization and dimensionality reduction algorithms like principal component analysis (Fig. 1b).

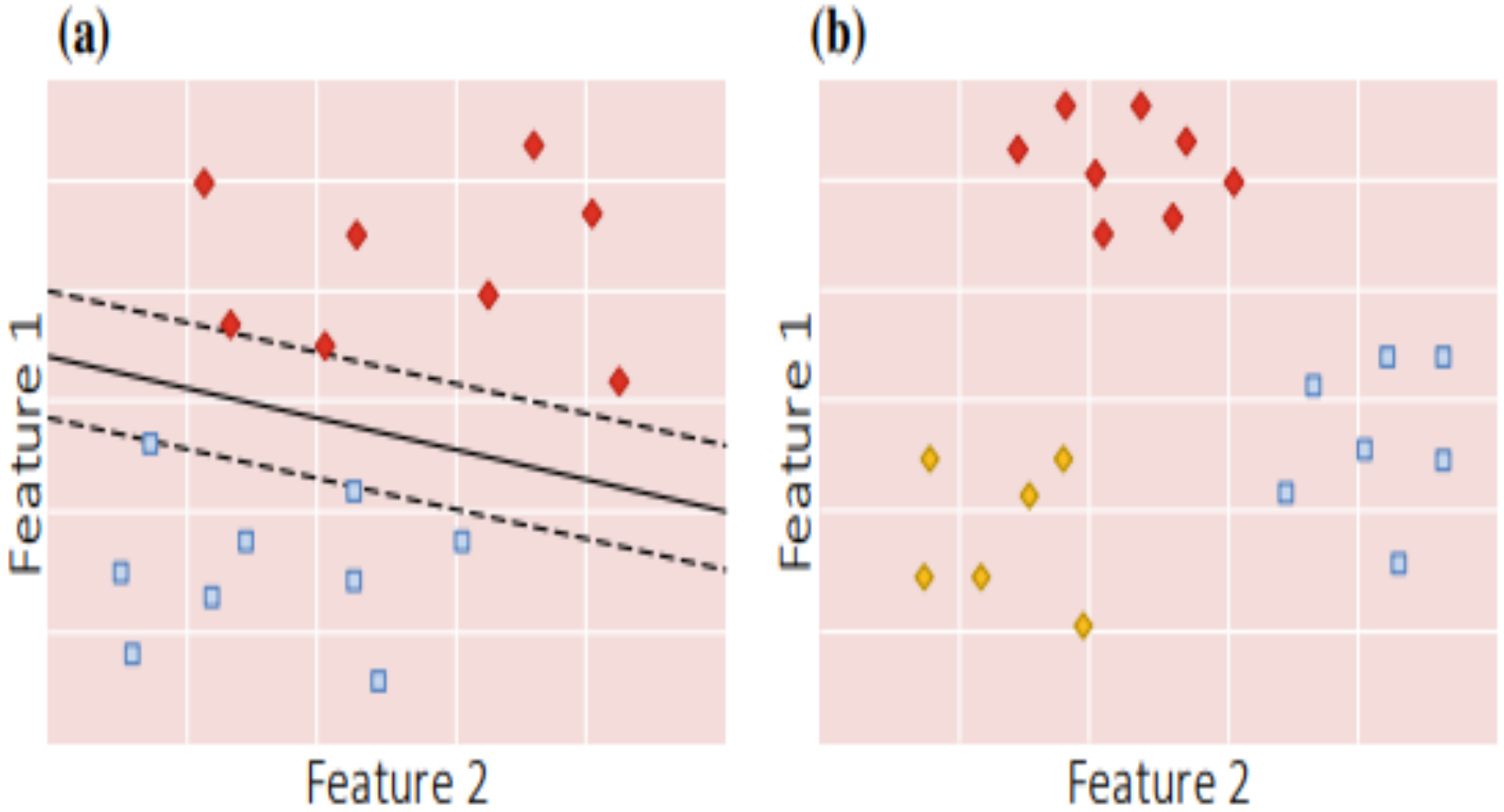


Fig. 1: (a) An example of a supervised learning algorithm, where a support vector machines technique was applied to classify the data into two different classes; (b) an example of an unsupervised learning algorithm, where a k-means technique was used to classify data into three different classes

In semi-supervised learning, the data used to train the system are partially labeled. Usually, the training data have a lot of unlabeled data points and a few labeled ones.

Most semi-supervised learning algorithms are combinations of supervised and unsupervised learning algorithms.

For example, deep belief networks are based on unsupervised components called restricted Boltzmann machines that are trained in an unsupervised manner and then the whole system is fine-tuned using supervised learning techniques.

Reinforcement learning implies the existence of an agent that can observe the environment and interact with it by selecting and executing actions. The agent gets a reward based on the action it chose (the reward can be positive or negative).

It must learn what is the best strategy, called policy, to maximize the reward over time. A policy is a function that defines what action the agent should choose when it is in a given situation.

These systems are used in robots to teach them how to walk. Reinforcement learning was also used on DeepMind's AlphaGo program that beat the Go game world champion.

It learned its policies by analyzing millions of games and then playing against itself.

- **Reinforcement learning is learning what to do—how to map situations to actions—so as to maximize a numerical reward signal. The learner is not told which actions to take, but instead must discover which actions yield the most reward by trying them.**
- **In the most interesting and challenging cases, actions may affect not only the immediate reward but also the next situation and, through that, all subsequent rewards. These two characteristics—trial-and-error search and delayed reward—are the two most important distinguishing features of reinforcement learning.**

Reinforcement learning

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Overview about Artificial Neural Networks

- **The building block of ANNS is the artificial neuron (also referred to as node).**
- **The operations that each neuron performs are simple: It sums its weighted inputs, passes that results through a function called activation function (usually nonlinear) and outputs that value, as shown in Fig. 2.**

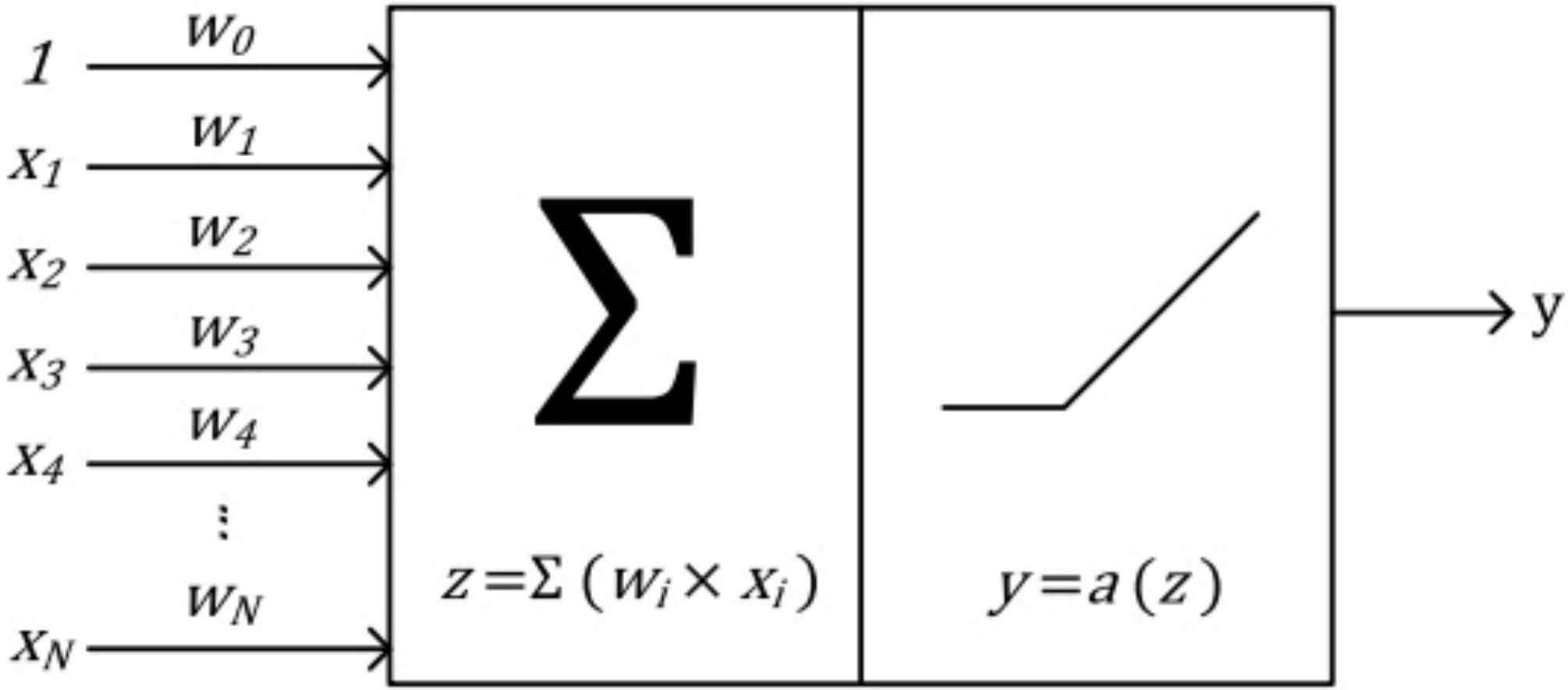


Fig. 2: Representation of an artificial neuron

Although these units execute simple operations, when they are grouped in dense networks, they can represent the most complex functions.

The parameters that define these functions are the weights of each neuron and are determined during training.

The fundamental structure for an artificial neural network is the multilayer perceptron (MLP) shown in Fig. 3. It consists of three or more layers of nodes (or neurons): One input layer, one output layer, and one or more hidden layers. The input layer is only responsible for distributing the inputs to the next layer: It does not do any operation with them.

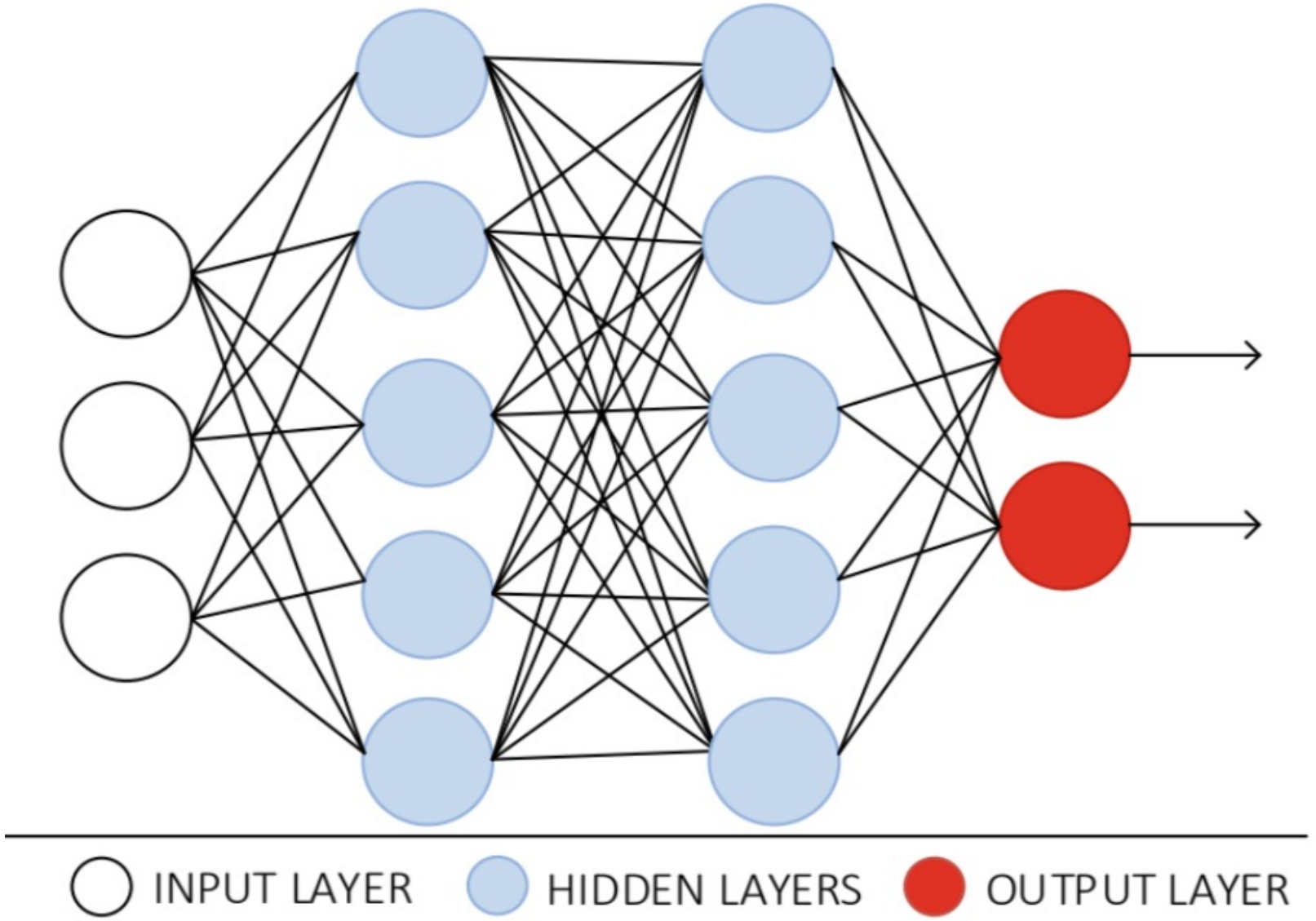


Fig. 3. Fully connected ANN with two hidden layers

The number of neurons in the input layer is always equal to the number of inputs of the model. The neurons in the hidden layers accept the outputs of the neurons of the previous layer and output them to the next layer.

These layers can have any number of neurons. The output layer receives the output of the last hidden layer, and its output corresponds to the output of the model.

Activation Functions

In the early days of ANNs, the most common activation functions are identity, sigmoid, and hyperbolic tangent, shown in Fig. 4.

Identity is used only at the output layer on regression problems so that the output of the model can be any value (positive or not).

In the hidden layers, however, the use of nonlinear functions makes it possible to represent any function using ANNs, and the sigmoid and the hyperbolic tangent started being used as the activation functions because they follow the behavior of biological neurons.

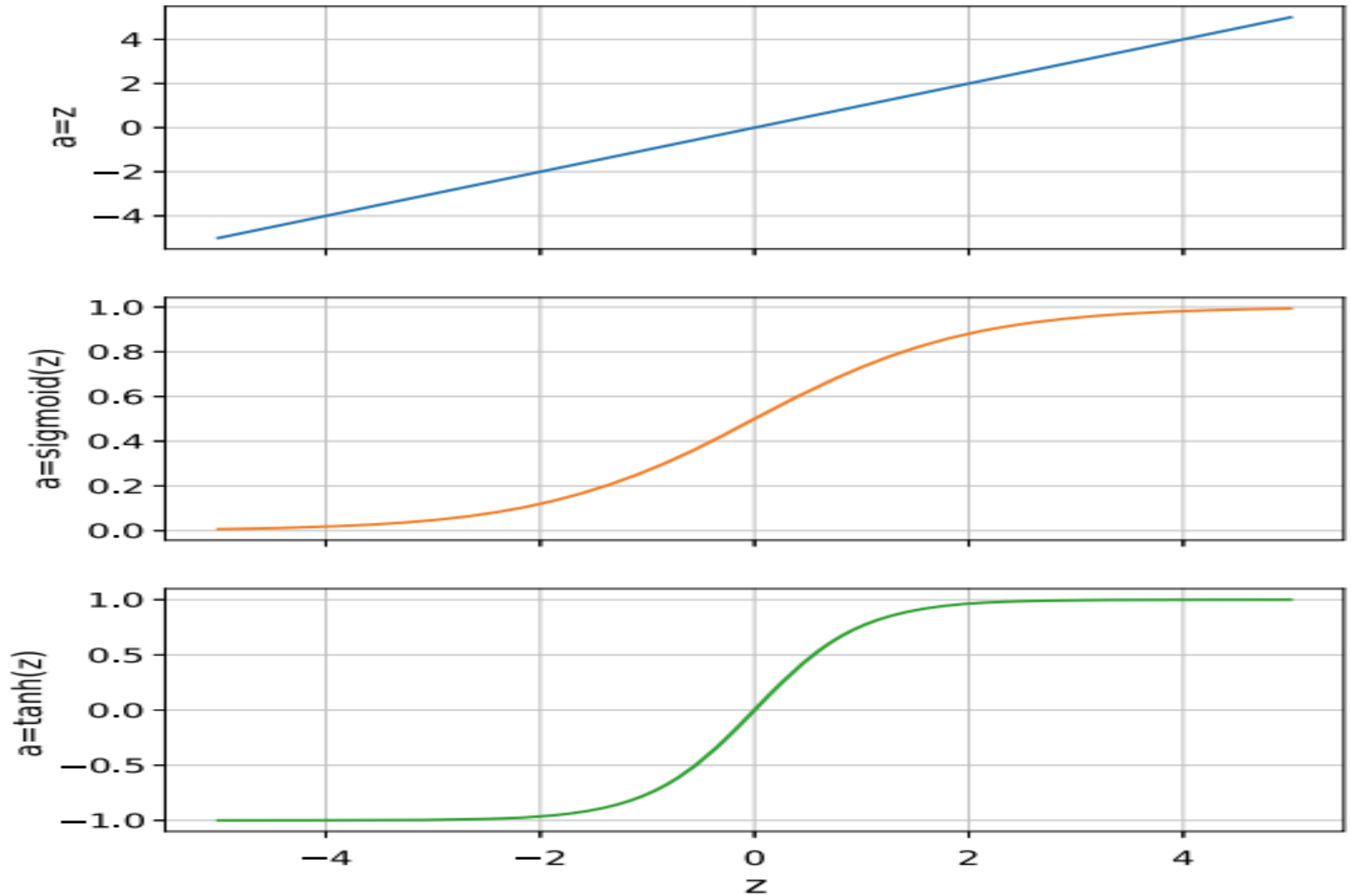


Fig. 4: Linear, sigmoid, and hyperbolic tangent functions

The rectified linear unit (ReLU) activation, as described in (1). The ReLU works better in deep networks because it does not saturate for positive values; besides, both function and gradient are very easy to compute.

$$\mathrm{ReLU}(z) = \max(0, z) \tag{1}$$

Still, once the sum of the weighted inputs of a neuron becomes negative, the neuron will output 0 with a gradient also of zero, which can lead to an average gradient of zero easily, once that happens those weights are no longer updated, leading to a problem known as the dying ReLUs. Some variants of the ReLU were proposed to address this problem. The leaky ReLU (lReLU) is defined according to (2).

$$lReLU(z) = \max(\alpha z, z) \tag{2}$$

The parameter α controls the slope (leak) of the function for z < 0, and its value is usually close to 0. α = 0.01 is considered a small leak, and α = 0.2 is considered a huge leak. The randomized leaky ReLU acts as the leaky ReLU, but the parameter α is picked randomly in a given range at training time and set to the average value during testing.

The parametric leaky ReLU turns the parameter α into a parameter that can be learned during training using the backpropagation method.

A new activation function called exponential linear unit (ELU) that outperformed all the ReLU variants reducing the training time and achieving better performance on the tested problems. The ELU function is defined by (3). The hyper-parameter α defines the value that the function tends to when z becomes a very large negative number.

$$\mathrm{ELU}(z) = \begin{cases} \alpha(e^z - 1) & \text{if } z < 0 \\ z & \text{if } z \geq 0 \end{cases} \tag{3}$$

The ELU is similar to the leaky ReLU but it is smooth for all values of z (including around z = 0), which helps speed up training algorithms. While it eventually saturates when z < 0, it still mitigates the occurrence of vanishing gradients problem and has a nonzero gradient in the negative spectrum, avoiding the dying units problem.

The main drawback of the utilization of this activation function is that is slower to compute than the ReLU variants, which makes it slower at test time, even though it converges faster during training time.

Figure 5 shows the ReLU, lReLU, and ELU functions.

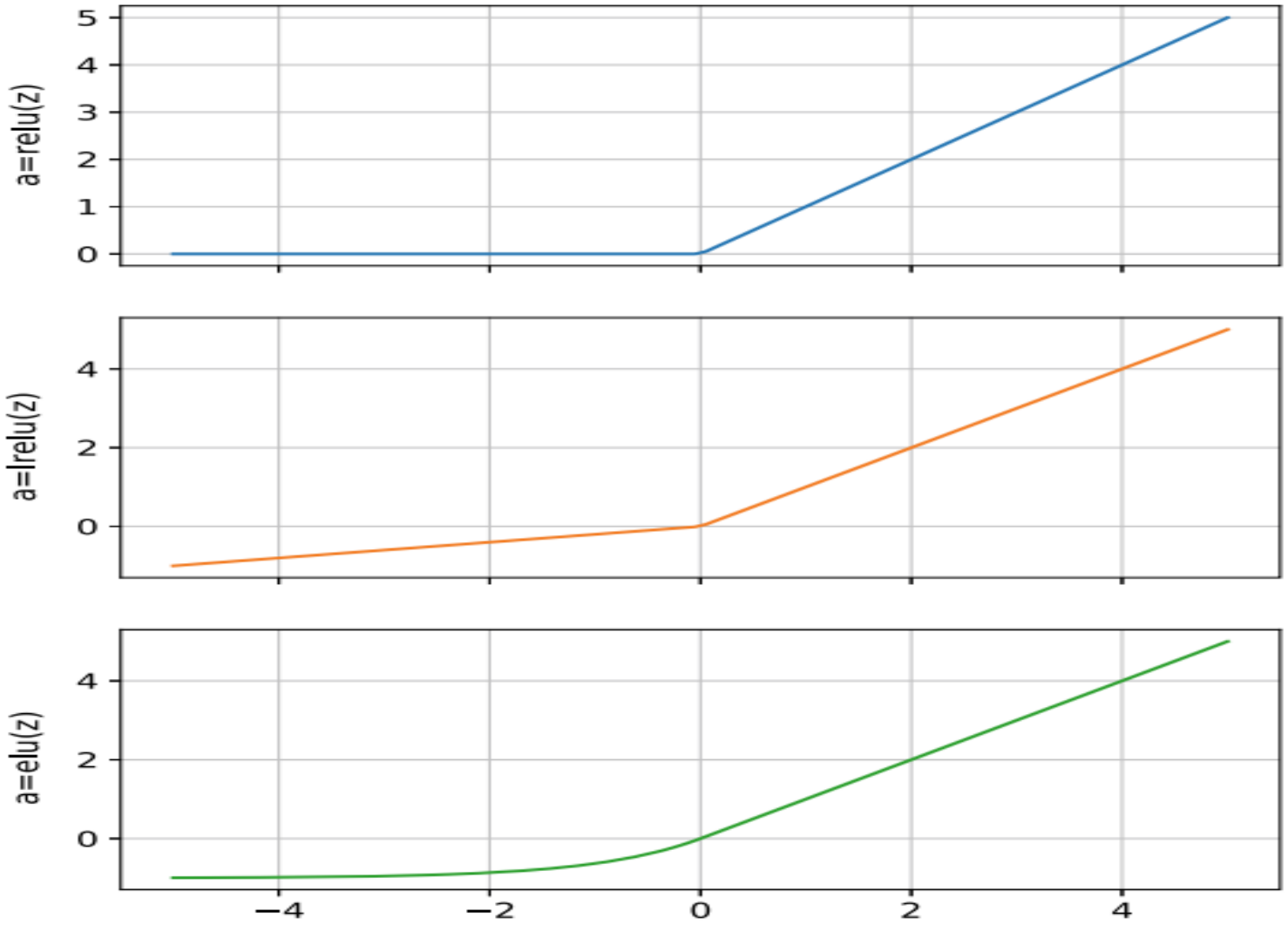


Fig. 5: ReLU, leaky ReLU for α = 0.2, ELU function for α = 1

H.W

- **In Reinforcement learning, What is the feature that distinguishes it from other types of learning?**
- **ML systems can be classified in broad categories, what are the factors which depend on?**
- **How can process the data in ANN? Describe it mathematically.**
- **Explain the types of activation functions with equations.**
- **Explain the three basic layers of ANN.**

References

- **Sutton, Richard S., and Andrew G. Barto. Reinforcement learning: An introduction. MIT press, 2018.**
- **Rosa, João PS, et al. "Overview of artificial neural networks." Using artificial neural networks for analog integrated circuit design automation (2020): 21-44.**